PHCP Reference Manual

The .phc model language, command by command

PHCP 1.5.0+ (v1.5.0-12-gfe1b4bd) - Perl Hydraulic Calculation Program

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The `.phc` model file, command by command, as PHCP reads it on the branch this manual is in.

This manual is the reference for every command, field, unit and option the `.phc` reader accepts. The README explains how PHCP works and what its report contains. This manual explains how to write a model.

`tools/test_manual.pl` checks the manual against the program, so it cannot fall behind. It confirms that every command, option, unit and fitting code the reader knows is described here, and that every complete example in this manual reads and solves.

1. The file

A `.phc` file is plain text. The program reads a file with Windows or Unix line endings alike.

One command to a line. The first word is the command. The fields follow it, separated by spaces or tabs. Fields go by position: there is no `KEY=value` form, and a field containing `=` is refused.

```
PIPE  P1  A  B  6000  50  2E T
```

Comments. `//` starts a comment that runs to the end of the line. A line may be all comment, or a command followed by one.

```
// RISER 1, LEVEL 3
PIPE  P1  A  B  6000  50     // DROP TO THE FIRST BRANCH
```

`!` and `#` are refused anywhere outside a comment.

The isometric drawing. The editor's ISOMETRIC tab stores each point's place on its grid in a comment, which this reader skips like any other comment. `//ISO GRID 1` is the metres of one grid step. `//ISO N1 0 0 4.5` is the point `N1`, the two plan steps and the vertical step. Deleting the comments leaves the calculation unchanged.

```
//ISO GRID 1
//ISO N1 0 0 0
//ISO N2 6 0 4.5
```

Case. Command words, units, method names, option names and pipe types may be typed in any case. `pipe`, `Pipe` and `PIPE` are the same command. Names of pipes, nodes, outlets and areas are case-sensitive: `S1` and `s1` are two different nodes.

Order. Lines may come in any order, with two exceptions:

- **USE** applies to the PIPE lines below it, up to the next USE.
- **Settings are file-wide.** TITLE, PROJECT, UNITS, METHOD and OPTION apply to the whole file wherever they appear. Each may be given only once: a second one is an error that names both lines.

Every node is defined exactly once. A node is defined by NODE, WATER, SPRINKLER, HYDRANT, HREEL or NC. A node that a pipe, pump or fixed flow names but nothing defines is an error. It names the node and the first line that uses it.

Line numbers. Every error the reader reports carries the line it comes from: `line 14: PIPE P7: 47 IS NOT A NOMINAL SIZE ...`. This includes the hydraulic checks made once the network is built.

Blank lines are ignored.

2. Values

Numbers

Numbers are plain decimals, optionally signed and with an exponent: `12`, `0.5`, `.5`, `-50`, `1.5e3`. A thousands separator is not allowed. `1,500` is not a number.

Lengths and elevations

A plain number is in the file's LENGTH or ELEVATION unit (see Section 3). A length may instead carry its own unit, written straight after the number:

written	means
`12M`	12 m
`300MM`	300 mm
`2.5CM`	2.5 cm
`6IN`	6 in
`10FT`	10 ft
`14'3"` or `14'3`	14 ft 3 in
`6'`	6 ft
`3"`	3 in

A pipe length cannot be negative. An elevation may be negative (below datum).

Sizes

The SIZE field of a pipe means one of two things, depending on the `USE` line in force:

under	SIZE is	in SIZE unit MM	in SIZE unit IN
`USE <pipe type>`	the nominal size	the DN: `50`, `100`	the nominal inch size: `2`, `1.5`, `4`
`USE BORE`	the internal bore	millimetres: `52.5`	inches: `2.067`

Nominal inch sizes accepted, and the DN each one means:

in	0.5	0.75	1	1.25	1.5	2	2.5	3	3.5	4	5	6
DN	15	20	25	32	40	50	65	80	90	100	125	150

in	8	10	12	14	16	18	20	24
DN	200	250	300	350	400	450	500	600

A size the pipe type is not made in is an error on its line. For example, `ASAM DN47` gives DATA_REQUIRED.

Names

A name is any run of characters without spaces: `P1`, `4101`, `RISER-2`, `L3.BR4`. Two restrictions apply:

- `END` cannot be a name. It is reserved.
- `~` cannot appear in a name. PHCP uses it for names it makes itself (see Section 12).

A pipe name is unique among pipes. A node name is unique among nodes. FIXEDFLOW and PUMP names share the pipe names: a pipe and a pump cannot have the same name.

Ratios

A pressure ratio is written `3:1`, `3:2` or as a number, such as `1.5`.

3. Units

Each quantity has its own unit, set by one `UNITS` line. A quantity that is not set takes the office default.

quantity	used for	units	default
`LENGTH`	pipe lengths, EQV lengths	`MM CM M IN FT`	`MM`
`ELEVATION`	node elevations	`MM CM M IN FT`	`M`
`SIZE`	nominal sizes, bores, orifice bores, roughness	`MM IN`	`MM`
`PRESSURE`	every pressure	`KPA BAR MPA PSI M` (metres head of water)	`KPA`
`FLOW`	every flow	`LPM LPS M3H GPM`	`LPM`
`KFACTOR`	outlet K-factors	`LPM/KPA LPM/BAR LPS/KPA LPS/BAR GPM/PSI`	`LPM/KPA`

These alternative spellings are also accepted:

typed	read as
`L/MIN`	`LPM`

typed	read as
L/S	LPS
M3/H	M3H
USGPM	GPM
L/MIN/KPA	LPM/KPA
L/MIN/BAR	LPM/BAR

The K-factor unit. K is flow divided by the square root of pressure, so the same head has a different K in each unit. A K80 head (80 L/min at 1 bar) is:

KFACTOR unit	K
LPM/BAR	80
LPM/KPA (the default)	8
GPM/PSI	5.6

The KFACTOR unit is independent of the FLOW and PRESSURE units. PHCP converts between them.

The report uses the file's FLOW and PRESSURE units, and gives velocity in m/s.

4. Command summary

The commands as the editor's command strip groups them. A short form, where there is one, is in brackets.

command	fields	section		
TITLE	TEXT	5.1		
PROJECT	FIELD TEXT	5.2		
UNITS	QUANTITY UNIT	5.3		
METHOD	HW or DW	5.4		
OPTION	NAME VALUE	5.5		
USE (U)	PIPETYPE [C]	6.1		
PIPE (P)	NAME START END LENGTH SIZE [FITTINGS ...]	6.2		
PC	SOURCE NAME START END [NAME START END ...]	6.3		
CHANGEPIPE (CHP)	PIPE LENGTH SIZE [FITTINGS ...]	6.4		
EQV	PIPE LENGTH	6.5		
NODE (N)	NAME ELEVATION	7.1		
NC	SOURCE NAME [NAME ...]	7.2		
WATER (W)	NAME ELEVATION STATIC [FLOW PRESSURE ...]	7.3		
SPRINKLER	NAME ELEVATION MINFLOW K	7.4		
HYDRANT	NAME ELEVATION MINFLOW K	7.4		
HREEL	NAME ELEVATION MINFLOW K	7.4		
FIXEDFLOW	NAME NODE FLOW [MINPRESSURE]	7.5		
NRV	SUPPLY CV [CRACK]	7.6		
NRVMODEL	SUPPLY TEXT	7.6		
PUMP	NAME IN OUT CHURN FLOW PRESSURE [FLOW PRESSURE ...]	8.1		
PUMPDUTY	PUMP FLOW PRESSURE	8.2		
BP	PIPE CHURN FLOW PRESSURE [FLOW PRESSURE ...]	8.3		
BFP	PIPE FLOW LOSS [FLOW LOSS ...]	8.4		
PRV	PIPE FLOW LOSS [FLOW LOSS ...]	8.4		
OPP	PIPE DROP [CD]	8.5		
OFD	PIPE BORE [CD]	8.5		
PRA	PIPE RATIO [MININLET] [BAND]	8.6		
DPV	PIPE SIZE RATIO\	TRIP SETTING ON\	OFF [FLOW LOSS ...] or [EQL LENGTH]	8.7
DPVMODEL	PIPE TEXT	8.7		
DPVAIR	PIPE PRESSURE	8.7		
DPVTRIP	PIPE ORIFICE [COUNT] [AIRTEMP] [STATIC]	8.7		
AREA	NAME OUTLET [OUTLET ...]	9.1		
SUPPLYGRAPH	FILE	9.2		

command	fields	section		
SCENARIO	NAME SUPPLY PERCENT [SUPPLY PERCENT ...]	9.3		

In the syntax lines below, <FIELD> must be given and [FIELD] may be left off. Optional fields go in order: to give the second, give the first.

5. Settings

Settings are file-wide and each is given once. In the editor, settings are made in the PROJECT tab rather than typed.

5.1 TITLE

```
TITLE <TEXT>
```

The job name, printed on the cover sheet and as the PDF's title. The text is the rest of the line, spaces included.

```
TITLE WAREHOUSE EXTENSION - STAGE 2
```

5.2 PROJECT

```
PROJECT <FIELD> <TEXT>
```

One cover-sheet field. The text is the rest of the line. Each field may be given once. The cover sheet uses:

field		field	
PROJECT	project name	STANDARD	design standard
JOB	job number	HAZARD	hazard class
COMPANY		DENSITY	design density, mm/min
ADDRESS		AREA_PER_SPRINKLER	m2
LICENCE		AREA_OF_OPERATION	m2
LOCATION		BASE_OF_RISER	
DESIGNER		AUTHORITY	approving authority
CHECKER		AUTHORITY_PHONE	
REVISION		DATA_SOURCE	where the water supply data came from
DATE			

With both DENSITY and AREA_PER_SPRINKLER given, the cover sheet checks the density each sprinkler delivers. Other field names are accepted and kept, but nothing prints them.

```
PROJECT JOB      HC-2026-041
PROJECT DESIGNER J. SMITH
PROJECT DENSITY  10.2
```

5.3 UNITS

```
UNITS <QUANTITY> <UNIT>
```

Sets one quantity's unit for the whole file (see Section 3). Each quantity may be set once.

```
UNITS LENGTH M
UNITS FLOW   LPS
UNITS KFACTOR LPM/BAR
```

5.4 METHOD

```
METHOD <HW|DW>
```

The friction method for every pipe:

value	method
HW (default; also HAZEN)	Hazen-Williams, to AS 2118.1:2017 Clause 14.10. Each pipe uses its C.
DW (also DARCY)	Darcy-Weisbach with the Colebrook-White friction factor. Each pipe uses its absolute roughness.

Under DW, a pipe's roughness comes from its USE line or, if that line gives none, from the database for the pipe type. A `USE BORE` pipe with no roughness is given 0.045 mm. That value is ASSUMED, and the report flags it in its warnings.

5.5 OPTION

```
OPTION <NAME> <VALUE>
```

A solver or design setting. Each option may be given once. An unknown option name is an error. A numeric option's value must be a number.

option	value	default	meaning
`HOSE_ALLOWANCE`	flow, FLOW unit	none	A hose stream flow added at the supply. The demand is checked at the system's flow plus this. It cannot be negative.
`DESIGN_AREA`	an AREA name	none	Solve and report that one design area only (Section 9.1).
`SIZING_GOAL`	`SMALLEST` or `VOLUME`	`SMALLEST`	For pipes given size 0 (Section 6.6): the smallest pipes that will do, or the least water held in the sized pipes.
`SIZING_MARGIN`	a percentage (`10%`) or a pressure (`50`, PRESSURE unit)	0	For pipes given size 0: how far under the supply the sized system must be at the demand point.
`SIZING_TRIALS`	1 to 50	5	For pipes given size 0: how many trials the sizing search makes, each from a different start.
`FITTING_BASIS`	`CORRECTED` or `STANDARD`	`CORRECTED`	CORRECTED corrects fitting lengths for C and for the installed bore. STANDARD uses the AS 2118.1 values unmodified (the "published Standard alone" election), and the report says so.
`VELOCITY_PRESSURE`	`ON` or `OFF`	`OFF`	ON: in-line sprinklers discharge on the normal pressure, q = K sqrt(Pt - Pv), Pv = rho v^2 / 2 in the pipe bringing the most water into the head's node (AS 2118.1 Clause 14.15, at the designer's discretion). End sprinklers, hydrants, hose reels and fixed flows stay on total pressure. OFF: every outlet on total pressure. The report and the JSON state which (Section 5.5.1).
`VEL_LIMIT`	m/s, above 0	6	Given in the model, a design limit: a pipe faster than it fails the design (NONCOMPLIANT). Left at the default of 6 m/s, a pipe faster than that is a warning only. A setting, not a clause of a Standard, and the report says which.
`SUPPLY_MARGIN`	pressure, PRESSURE unit	0	The least margin the supply must have over the demand. With one supply, at the demand point; with several, at their joint demand (Section 9.3). Less fails the design. It cannot be negative.
`SUPPLY_SCENARIOS`	`ON` or `OFF`	`ON`	With more than one supply: ON solves and reports, for information, each supply alone and each out of service in turn. SCENARIO lines are solved either way (Section 9.3).
`JOINT_BASIS`	`DERATE` or `SCALE`	`DERATE`	With more than one supply, how they are reduced together to find their joint demand: every supply's pressure lowered by one common amount (DERATE), or multiplied by one common factor (SCALE). An assumption, and the report states it (Section 9.3).
`HW_CONST`	number, within 2 % of 6.05e7	6.05e7	The Hazen-Williams constant: P = k Q^1.85 / (C^1.85 d^4.87), P in kPa/m, Q in L/min, d in mm. Its published forms (AS 2118.1 and EN 12845 6.05 x 10^5 bar/m; NFPA 13 4.52 in US units, 6.049 x 10^7 converted) lie within 0.6 % of 6.05e7, so anything further than 2 % is refused: a constant for other units, such as 6.05E5 or 4.52, would change every friction loss.

option	value	default	meaning
NU	m2/s, 0.28e-6 to 1.8e-6	1.004e-6	Kinematic viscosity of water, for Darcy-Weisbach. The default is water at 20 C; the range is water from 100 C to 0 C, and a figure outside it is refused.
TOL_FLOW	flow, FLOW unit, above 0	0.05 L/min	Convergence tolerance on flow: when the iteration may stop.
TOL_LOOP	pressure, PRESSURE unit, above 0	0.01 kPa	Convergence tolerance on each loop's pressure balance: when the iteration may stop. It is not the test of the answer: every loop and every element equation is checked afterwards to PHCP's verification limit, 0.1 kPa, whatever TOL_LOOP is, and a solve that stopped further out is REJECTED.
MAXITER	count	50000	The most Hardy Cross iterations. A solve that has not converged by then is reported NOT CONVERGED.
ORIF_STEP	mm, above 0	0.5	The step an orifice plate sized by OPP is rounded to.
ORIF_DP_TOL	per cent, above 0	10	How far an OPP plate's final drop may be from its target. Further fails the design (NONCOMPLIANT, ORIFICE_DUTY). 10 % is what half a 0.5 mm step does to the drop of a 10 mm bore; a tighter tolerance needs a finer ORIF_STEP. The report gives each plate's deviation split into the part from rounding the bore and the part from the flow moving.
RELAX	above 0, at most 2	none	Accepted and range-checked for compatibility. The present solver does not use it.

```
OPTION HOSE_ALLOWANCE 500
OPTION VEL_LIMIT      7.5
OPTION FITTING_BASIS  STANDARD
OPTION VELOCITY_PRESSURE ON
```

5.5.1 VELOCITY_PRESSURE

AS 2118.1:2017 Clause 14.15 lets the designer include velocity pressures and, where they are included, requires them for range pipes and distribution mains alike. It gives no method. With `OPTION VELOCITY_PRESSURE ON`:

- Only `SPRINKLER` outlets change. `HYDRANT`, `HREEL` and `FIXEDFLOW` stay on total pressure.
- A sprinkler with a pipe or device carrying water on beyond its node (an in-line head) discharges on Pn = Pt - Pv. Pv = rho v^2 / 2, 0.5 v^2 kPa, with v the velocity in the pipe bringing the most water into the node.
- A sprinkler with nothing carrying water beyond it (an end head: the end of a range, a drop or an armover) discharges on Pt.
- Every pipe is treated alike, so range pipes and distribution mains are both covered. No normal pressure is taken at a branch take-off; branches are solved in full.
- Pv is set from a converged solve and held while the network is solved again, until no Pv moves by more than `TOL_LOOP`.

Friction, elevation and node pressures stay total pressure. Outlet compliance, margins, the governing outlet and the demand point are measured on the pressure each outlet discharges on. The default, `OFF`, gives exactly the results PHCP gave before the option existed.

6. Pipes

6.1 USE

```
USE <PIPETYPE> [C]
```

Sets the pipe type for the PIPE lines below it, up to the next USE.

field	meaning
PIPETYPE	A pipe type code from the database (Section 11), or `BORE` for pipes given by their bore.

field	meaning
C	Under METHOD HW, the Hazen-Williams C. Under METHOD DW, the absolute roughness, in the SIZE unit. If it is left off, the database's value for the pipe type is used.

Before the first USE line, pipes are `USE BORE 120`.

Fittings need a pipe type, because their equivalent lengths come from the database by type and size. A pipe under `USE BORE` cannot list fittings: give its allowance with EQV instead.

```
USE ASAM            // AS 1074 MEDIUM, C FROM THE DATABASE (120)
USE S20 110         // SCHEDULE 20 AT C 110
USE BORE 100        // PIPES GIVEN BY BORE, C 100
```

6.2 PIPE

```
PIPE <NAME> <START> <END> <LENGTH> <SIZE> [FITTINGS ...]
```

field	meaning
NAME	The pipe's name, unique among pipes.
START, END	The nodes at each end. Positive flow runs START to END, and the report signs flows that way.
LENGTH	The pipe's length, LENGTH unit (or with its own unit).
SIZE	The nominal size, or under USE BORE the bore (Section 2).
FITTINGS	Fitting codes, each with an optional count: `2E T GV`. See Section 10.

The pipe takes the type and C of the USE line above it.

```
USE  ASAM
PIPE P1  W   R1  12000  100  E GV C
PIPE P2  R1  J1  3500   80   2E T
```

6.3 PC - copy a pipe

```
PC <SOURCE> <NAME> <START> <END> [<NAME> <START> <END> ...]
```

Makes one or more pipes that are the same as SOURCE, with the same type, C, length, size and fittings, between new nodes. Each copy is three fields: NAME START END.

A copy takes its source as defined, before any CHANGEPIPE on the source. A source may itself be a copy. A chain of copies that comes back on itself is an error.

```
PIPE P10 J1 S1 3000 25 T
PC   P10 P11 J2 S2  P12 J3 S3  P13 J4 S4
```

6.4 CHANGEPIPE - change a pipe

```
CHANGEPIPE <PIPE> <LENGTH> <SIZE> [FITTINGS ...]
```

Replaces a pipe's length, size and fittings, keeping its name, nodes, type and C. It is normally used on a copy, to make it different from its source. The source keeps its original values. Each pipe may be changed once.

```
PC         P10 P14 J5 S5
CHANGEPIPE P14 4500 32 E T
```

6.5 EQV - add an equivalent length

```
EQV <PIPE> <LENGTH>
```

Adds a length to the pipe's equivalent length, on top of its fittings. It is the way to give a fitting allowance on a `USE BORE` pipe, or one the database does not hold. Each pipe takes one EQV.

```
USE  BORE 120
PIPE P5 A B 6000 52.5
EQV  P5 3.2M
```

6.6 Auto sizing: SIZE 0

A PIPE or CHANGEPIPE with SIZE `0`, under a `USE <pipe type>`, is sized by PHCP from the sizes the database has for that pipe type.

```
OPTION SIZING_GOAL   SMALLEST
OPTION SIZING_MARGIN 10%
USE    ASAM
PIPE   LA1 J1 A1 3000 0 T
```

The settings that control it:

setting	value	default	meaning
SIZING_GOAL	SMALLEST	yes	the smallest pipes that will do
	VOLUME		the least water held in the sized pipes (for a dry or preaction system)
SIZING_MARGIN	10%		at the demand point the system needs at least 10 % less than the supply gives
	50		...or at least 50 less, in the PRESSURE unit
	(not set)	0	no margin
SIZING_TRIALS	1 to 50	5	how many trials the search makes

A size "will do" when every outlet gets its requirement from the supply reduced by the margin, and no sized pipe runs faster than VEL_LIMIT (6 m/s unless set) at the operating point. With one supply this is checked at the demand point itself. With design areas (Section 9.1) every area must pass. Pipes not given size 0 are never changed.

What is sized together. A PIPE line sizes its pipe and every PC copy of it not changed since. A CHANGEPIPE line with size 0 sizes the one pipe it changes.

How. The search makes several trials (SIZING_TRIALS, 5 unless set), each from a different start, and keeps the best one that passes:

trial	starts from
1	every pipe at its smallest size
2	every pipe at its largest size
3 onwards	the best so far, with about a third of the pipes moved a size or two up and a third a size down

The variations follow a fixed sequence, so the same model always gives the same sizes. Each trial then works in three steps:

1. **Up:** the pipe whose next size up does the most good for its cost is enlarged, one step at a time, until the network passes.
2. **Down:** each pipe is tried one size smaller, keeping any that still pass.
3. **Swap:** one pipe is moved a size up and another a size down, keeping the change when it still passes and costs less.

"Best" means the least sum of length x bore over the sized pipes (SMALLEST), or the least water held in them (VOLUME). The report lists each trial and which one was chosen. More trials can find smaller pipes, but take longer. While it works, the editor's status bar shows COMPUTING, the trial, the number of solves and the seconds gone.

The answer.

- In the editor, CALCULATE writes each size after its line as a comment, leaving SIZE 0 so the pipe is sized again next time:

```
PIPE   LA1 J1 A1 3000 0 T  //AS - 40
```

A comment already on the line is kept after the size. An old `//AS` comment is replaced.

- On the command line, `phcp.pl model.phc --annotate out.phc` writes the model with the comments.
- The report's Section 1 lists every sized pipe, its size and velocity, and the margin met against the margin asked for. If even the largest sizes do not pass, the pipes are left at the largest, and the report and warnings say NOT MET.

Auto sizing needs a pipe type: SIZE 0 under `USE BORE` is an error.

7. Nodes and outlets

7.1 NODE

```
NODE <NAME> <ELEVATION>
```

A junction, at an elevation above the datum, in the ELEVATION unit. All elevations in a file share one datum.

```
NODE R1  0
NODE J1  6.5
```

7.2 NC - copy a node

```
NC <SOURCE> <NAME> [NAME ...]
```

Makes nodes that are the same as SOURCE: the same elevation and, for an outlet, the same kind, MINFLOW and K. A copy of a SPRINKLER is a sprinkler.

```
SPRINKLER S1 6.5 -50 8
NC        S1 S2 S3 S4 S5
```

7.3 WATER - a water supply

```
WATER <NAME> <ELEVATION> <STATIC> [FLOW PRESSURE ...]
```

A supply connection, defining its node. A model may have any number of supplies.

field	meaning
ELEVATION	The elevation the pressures are measured at.
STATIC	The pressure at no flow.
FLOW PRESSURE	Further points on the supply curve: residual pressure at each flow.

With no pairs, the supply holds STATIC at any flow, like a tank or a fixed head. With pairs, the supply is a curve from STATIC at no flow through every point given. Between points, pressure is a straight line on Q^1.85 paper. Beyond the last point, the curve continues in a straight line in flow.

```
WATER TOWN 0 650 1800 560 3600 400
WATER TANK 12 0
```

With one supply, the report finds the demand point: the supply pressure at which the governing outlet has exactly its requirement.

7.4 SPRINKLER, HYDRANT, HREEL - outlets

```
SPRINKLER <NAME> <ELEVATION> <MINFLOW> <K>
HYDRANT   <NAME> <ELEVATION> <MINFLOW> <K>
HREEL     <NAME> <ELEVATION> <MINFLOW> <K>
```

Defines the outlet's node and the outlet on it together. The kind matters to the report, which counts sprinklers apart from hydrants and hose reels.

field	meaning
ELEVATION	The outlet's elevation.
MINFLOW	The least the outlet must give. A positive value is a flow (FLOW unit). A negative value is instead a pressure (PRESSURE unit): `-50` means at least 50 kPa.
K	The K-factor, in the KFACTOR unit. Flow = K x sqrt(pressure). A K of 0 shuts the outlet: its node stays, discharging nothing.

Every outlet is checked, PASS or FAIL. The outlet with the least pressure to spare over its requirement is the governing outlet.

```
SPRINKLER S1  6.5  -50   8      // AT LEAST 50 KPA, K 8 L/MIN/KPA^0.5
SPRINKLER S2  6.5   61   8      // AT LEAST 61 L/MIN
HYDRANT   H1  0.5  600   21
HREEL     HR1 1.0  20    3
```

7.5 FIXEDFLOW - a fixed demand

```
FIXEDFLOW <NAME> <NODE> <FLOW> [MINPRESSURE]
```

A flow taken from a node whatever its pressure: a hose stream, a process draw, a flow measured elsewhere. It is a nodal demand, outside every loop. NAME shares the pipe names, and NODE must be defined elsewhere.

MINPRESSURE, when given, is the pressure needed to deliver the flow, and it is checked. The fixed flow is listed with the outlets, PASS or FAIL on pressure alone. A fixed flow with the least pressure to spare governs like any outlet. Its flow is never reduced when pressure is short: the report shows the shortfall as a FAIL instead.

```
FIXEDFLOW HOSE1 J4 450 250
```

7.6 NRV, NRVMODEL - a check valve at a supply

```
NRV      <SUPPLY> <CV> [CRACK]
NRVMODEL <SUPPLY> <TEXT>
```

A non-return (check) valve at a water supply. Water may pass from the supply into the network, never back. A supply has no check valve unless an NRV line gives it one: none is assumed.

field	meaning
SUPPLY	A water supply (WATER, or a copy of one). One NRV per supply.
CV	The valve's flow coefficient from its data sheet, in US units: the loss in psi is (flow in gpm / CV) squared. 0 says the valve's flow loss is already in the pipework, for example as an EQV on the supply's pipe.
CRACK	The pressure it takes to open the valve. Taken as 0 when not given, and the report says so.
TEXT	NRVMODEL: the make and model, as typed. The data sheet the CV and CRACK came from.

When open, the valve's loss is CRACK plus the CV loss at the flow through it.

The program decides each valve's state from the solved network, after every Hardy Cross solve that converges, and solves again when any valve changes:

- An open valve whose supply would take water in is CLOSED. The supply is taken out of the network and the network is solved without it.
- A closed valve OPENS again when the supply's pressure at no flow, plus its elevation, exceeds the network at its connection by more than CRACK.

With several supplies, a weak supply shuts and the others carry the load. It may open again as the demand rises, for example in a demand search. A supply shut on its valve is CLOSED, a valid operating state. It is not a supply found too weak, and not an invalid trial.

A valve is never held open to keep the network connected. If every supply that could feed part of the network is shut, that part has LOST ITS SUPPLY. It is shown with no flow and no pressure, the outlets there are named, and the calculation is NONCOMPLIANT (SUPPLY_ISOLATED). This holds for a single supply too: a supply that cannot lift water to its outlets shuts, and the model is not accepted. Without a check valve the same model is REJECTED, because the supply would take water in.

A valve that keeps opening and closing on alternate solves is stopped after 8 changes. If it is still in the wrong state, the calculation is INDETERMINATE (SUPPLY_CHECK_UNRESOLVED).

```
WATER    TOWN 0 650 1800 560 3600 400
NRV      TOWN 950 7
NRVMODEL TOWN ACME 150 MM SWING CHECK, DATA SHEET 12
```

Section 10 of the report lists every supply's valve, state, flow, pressure and loss. The JSON gives the same under `supply_check_valves`.

8. Pumps and devices

8.1 PUMP

```
PUMP <NAME> <IN> <OUT> <CHURN> <FLOW> <PRESSURE> [FLOW PRESSURE ...]
```

A pump between two nodes, adding pressure from IN to OUT. CHURN is the boost at no flow. Each FLOW PRESSURE pair is a further point on its curve, read as a supply curve is (Section 7.3). At least one pair is required. NAME shares the pipe names.

A pump does not run backwards. If the network would push water from OUT to IN, the pump's check valve closes it.

```
PUMP FP1 SUC DIS 700 2000 600 3000 450
```

8.2 PUMPDUTY - a pump's rated duty

```
PUMPDUTY <PUMP> <FLOW> <PRESSURE>
```

The rated duty of a PUMP, given by its name, or of a booster in a pipe (BP), given by the pipe's name. The report's PUMP CURVES section reads the curve against it:

check	rule	counts as
AT DUTY	the curve at the duty flow gives at least the duty pressure	a check; a failure is a warning
SHUT-OFF	the curve at no flow is at most 140 % of the duty pressure (AS 2941-2013, from the Pump Industry Australia fire pump checklist)	a check; a failure is a warning
150 % FLOW	the curve at 150 % of the duty flow gives at least 65 % of the duty pressure (NFPA 20, 6.2)	for information only, never a failure

A reading beyond the curve's last point is marked EXTRAPOLATED. Each pump takes one duty. A pump without a duty is listed with its shut-off pressure only.

```
PUMPDUTY FP1 2000 600
```

8.3 BP - a booster pump in a pipe

```
BP <PIPE> <CHURN> <FLOW> <PRESSURE> [FLOW PRESSURE ...]
```

A pump in a pipe, at the pipe's start. The fields are as for PUMP after the nodes. The device is named `<pipe>.BP` in the report (Section 12).

```
BP P1 300 1500 250 2500 180
```

8.4 BFP, PRV - a backflow preventer or pressure reducing valve

```
BFP <PIPE> <FLOW> <LOSS> [FLOW LOSS ...]
PRV <PIPE> <FLOW> <LOSS> [FLOW LOSS ...]
```

A device in a pipe, at the pipe's start, that loses pressure with flow along the curve given: LOSS at each FLOW, in the PRESSURE and FLOW units, read as a supply curve is.

A single pair at flow 0 is a FIXED loss that does not vary with flow: `BFP P1 0 35` is 35 kPa at any flow. A single pair at any other flow is an error, and a fixed loss must be more than 0.

```
BFP P1 0 10 2000 40 4000 75
PRV P9 0 50
```

8.5 OPP, OFD - orifice plates

```
OPP <PIPE> <DROP> [CD]
OFD <PIPE> <BORE> [CD]
```

An orifice plate in a pipe, at the pipe's start. It takes its pipe's nominal size and bore.

command	given	found
OPP	the DROP wanted, PRESSURE unit	the plate's bore, rounded to ORIF_STEP (0.5 mm unless set)
OFD	the plate's BORE, SIZE unit	the drop at the solved flow

CD is the discharge coefficient, 0.596 unless given; it must be more than 0 and not more than 1. The drop is the net loss across the plate: the differential across it, 224.954/CD^2 x Q^2/d^4 x (1 - b^4) kPa (Q in L/min, d the bore in mm, b = d/D), times the ISO 5167 permanent-loss ratio, which also depends on CD. At a fixed bore the differential goes as 1/CD^2, so a plate given CD 0.7826 has (0.596/0.7826)^2 = 0.58 of the differential it has at 0.596, and OPP finds a smaller bore for the same drop.

```
OPP P7 120
OFD P8 38.5 0.61
```

8.6 PRA - a pressure ratio valve

```
PRA <PIPE> <RATIO> [MININLET] [BAND]
```

A pressure ratio valve in a pipe, at its start. It holds its outlet at its inlet pressure divided by RATIO.

field	meaning
RATIO	`3:1`, `2:1`, `1.5`, ... It must be more than 1: a ratio valve reduces pressure.
MININLET	The manufacturer's least inlet pressure. An inlet below it is a warning. 0 means no minimum, so a BAND can be given without one.
BAND	The valve's tolerance, in % of the inlet pressure. 5 if not given. 0 means the nominal ratio exactly.

The calculation uses the BOTTOM of the band, outlet = inlet x (1/ratio - band), so every outlet is checked against the least pressure the valve can be relied on to give. The report solves the network once more with every valve at the TOP of its band, and states the highest pressure that gives downstream. A band wider than the ratio passes is refused.

```
PRA P2 3:1 300
PRA P3 2:1 0 3
```

8.7 DPV, DPVMODEL, DPVAIR, DPVTRIP - a dry pipe valve

```
DPV <PIPE> <SIZE> RATIO <RATIO> <ON|OFF> [FLOW LOSS ...]
DPV <PIPE> <SIZE> TRIP <PRESSURE> <ON|OFF> [FLOW LOSS ...]
DPV <PIPE> <SIZE> RATIO <RATIO> <ON|OFF> EQL <LENGTH>
DPV <PIPE> <SIZE> RATIO <RATIO> <ON|OFF> VALVE <MAKER> <MODEL>
DPVMODEL <PIPE> <TEXT>
DPVAIR <PIPE> <PRESSURE>
DPVTRIP <PIPE> <ORIFICE> [COUNT] [AIRTEMP] [STATIC]
```

A dry pipe valve in a pipe, at its start. PHCP solves it TRIPPED AND FULLY OPEN: the steady state after the valve has operated and the system has filled. It calculates no trip time, transit time or water delivery time, and none of its results says one has been checked.

field	meaning
SIZE	The valve's nominal size, in the SIZE unit (100, or 4 under UNITS SIZE IN). 25 to 300 mm.
RATIO or TRIP	How the valve's trip is set. RATIO: a differential trip ratio, the water pressure to the air pressure at which the clapper is held, `5.5:1` or `5.5`. It must be more than 1:1, and one outside 3:1 to 8:1 is a warning to check. TRIP: the air-side pressure at which the valve trips, in the PRESSURE unit, above 0. Give one, not both. A trip ratio is not a loss coefficient.
ON or OFF	The accelerator. No effect on trip time is assumed or calculated.
FLOW LOSS ...	The manufacturer's loss through the open valve: flow and loss pairs, read as every PHCP curve is, straight in Q^1.85 between the points and continued beyond the last (which is a warning). One pair at flow 0 is a loss that does not vary with flow.
EQL LENGTH	Instead, the valve's equivalent length of its pipe, in the LENGTH unit, worked at that pipe's bore and C (or roughness).
VALVE MAKER MODEL	Instead, a manufacturer's valve from the pipe and fitting database (`DRYVALVE` records, `data/pipes.dat`), in the size SIZE: its published equivalent length of its published reference pipe and C, worked by Hazen-Williams. The database holds Reliable, Tyco, Viking and Victaulic valves; an unknown valve, or a size it is not made in, is refused with the ones there are. The data sheet's trip ratio and rated working pressure are checked against the model: a ratio far from it, a low-pressure valve given a ratio, or a legacy valve is a warning; an inlet pressure above the rating REJECTS the calculation. A conflict found in the data sheet itself is printed with the valve.

The loss must be given. A DPV with no loss data is solved with none and the calculation is REJECTED: no loss is ever assumed for it, and a check valve's is not put in its place.

The trip setting and the accelerator do not change any steady-state figure. DPVMODEL is the make and model, free text, the rest of the line. DPVAIR is the air pressure the system is set to.

Estimated trip, transit and delivery times. With DPVAIR and DPVTRIP the report gives, for the valve, ESTIMATES - they change no other figure, do not decide the calculation's acceptance, and are not a water delivery time calculation by a listed method:

- TRIP TIME by the formula of FM Global, as published in the SFPE Handbook: t = 0.0352 V / (An sqrt T0) ln(pa0 / pa), V the dry volume (every pipe downstream of the valve), An the open orifice area, T0 the air temperature, pa0 the set air pressure and pa the trip pressure (TRIP, or the static water pressure divided by RATIO), all absolute.
- TRANSIT TIME, ignoring the air: the larger of the water front's travel time from the valve to the most remote outlet, the sum of L / v along the flow path, and the time to fill the dry volume at the flow through the valve, both at the demand point's flows (with one supply) or else the operating point's. A lower bound: the air the water must push out is not modelled.
- DELIVERY TIME, trip plus transit.

The accelerator is not modelled: a time with ON is the time without it, and the report says so.

DPVTRIP field	meaning
ORIFICE	The nominal orifice of the open sprinkler (or test outlet), in the SIZE unit.
COUNT	How many are open. 1 if not given.
AIRTEMP	The air temperature in the dry pipework, degrees C, -50 to 60. 20 if not given. Colder air gives a longer trip time.
STATIC	The static water pressure at the valve, in the PRESSURE unit, for a RATIO trip. If not given, the highest supply's pressure at no flow, corrected for elevation; pumps are not included.

A set air pressure not above the trip pressure is a valve that cannot be set, and REJECTS the calculation. A trip pressure below 1.893 atmospheres absolute, where flow through the orifice stops being choked, is a warning. A dry part that reaches a water supply is refused.

The report's Section 13 lists each valve: where it is, its size, make, trip, accelerator, loss basis, steady-state flow and loss, the pressures either side, and its checks. Water backwards through it, a size that is not a valve size, or a trip setting that cannot be right REJECTS the calculation.

```
DPV P12 100 RATIO 5.5:1 ON 1000 15 3000 40
DPV P14 150 TRIP 140 OFF EQL 6.1
DPV P16 100 RATIO 5.5:1 ON VALVE TYCO DPV-1
DPVMODEL P12 MAKER X MODEL D100
DPVAIR P12 280
DPVTRIP P12 12.7 1 5
```

One device to a pipe. BP, BFP, PRV, OPP, OFD, PRA and DPV name the pipe they sit in, and a pipe holds one device. A second one is an error naming the first.

9. Design and reporting

9.1 AREA - a design area

```
AREA <NAME> <OUTLET> [OUTLET ...]
```

Names outlets that flow together. A model may hold more outlets than ever flow at once (a whole floor, of which the design area is one part), and each AREA names one set. A second AREA line with the same name adds to that area.

Each OUTLET must be a SPRINKLER, HYDRANT or HREEL node, or a copy of one. An outlet named in no AREA, such as a hydrant or hose reel, flows in every area.

With AREA lines, PHCP solves each area on its own, with the other areas' outlets shut, and reports the one that governs:

4. an area whose solve did not converge, first;
5. else, with one supply, the area with the least margin at the supply;
6. else the area whose governing outlet has the least pressure to spare.

Section 1 of the report lists every area and marks the one that governs. To solve and report one area only, name it with `OPTION DESIGN_AREA`.

```
AREA NEAR A1 A2 A3 A4
AREA FAR  B1 B2 B3 B4
AREA MID  A3 A4
AREA MID  B1 B2
```

9.2 SUPPLYGRAPH - a graph sheet in the report

```
SUPPLYGRAPH <FILE>
```

Adds a water supply graph made in the editor's GRAPH tab (a `.phg` file) to the model's report:

- in the text report, an appendix with the graph's points and its duty points against the curve designed against;
- in the PDF, the sheet itself, on a page after the text.

FILE is the rest of the line, so a name with spaces is fine. A relative name is found from the model's folder, or the current folder for a model not yet saved. More than one may be given; they become appendices A, B and so on.

A graph that cannot be read, or that the GRAPH tab would refuse, is listed in the warnings, and the report is written without it.

```
SUPPLYGRAPH town main.phg
SUPPLYGRAPH C:\JOBS\HC-041\PUMP TEST.phg
```

9.3 SCENARIO - a water supply scenario

```
SCENARIO <NAME> <SUPPLY> <PERCENT> [SUPPLY PERCENT ...]
```

With more than one supply, a condition the design must also meet: each SUPPLY named runs at PERCENT of its pressure (its static, or every point of its curve), 0 to 100. 0 is out of service: the supply is taken out and its node stays as a junction. Supplies not named run as modelled.

Each scenario is solved and validated on its own (Section 14 of the report). A SCENARIO that is not ACCEPTED fails the design. With `OPTION SUPPLY_SCENARIOS ON`, the default, PHCP also solves each supply alone and each out of service in turn; those are reported for information only.

With more than one supply, the report's Section 12 gives, for each supply:

- its OPERATING flow and pressure, as modelled, its share, and whether it takes water in (which REJECTS the calculation);
- its CONDITIONAL MINIMUM: the least pressure at its connection, and the flow, for the governing outlet to receive exactly its minimum with every other supply as modelled. The conditional minima of different supplies are not a condition they can all be at together, and are never added;
- its flow and pressure at the JOINT DEMAND: one solve with every supply reduced together on the basis `OPTION JOINT_BASIS` names, until the governing outlet receives exactly its minimum. Each supply keeps its own curve and connection; their pressures are not assumed equal.

```
SCENARIO TOWNLOW TOWN 80
SCENARIO NOPUMP PUMPSET 0 TOWN 90
```

10. Fittings

Fittings follow a pipe's SIZE, or a CHANGEPIPE's. Each is a code with an optional count in front: `E`, `2E`, `T`, `3GV`. They may also be run together or comma-separated, `2E,1T` or `2E1T`, as written on many calculation sheets. A count must be 1 to 99.

Short forms.

typed	read as
`G`	`GV`
`C`	`CV`
`B`	`BV`

Which table applies. Each length comes from the database, by the pipe's type and size:

pipe type	table	correction
ASAM, ASAH, S20, S40, DN150 and below	AS 2118.1:2017 Table 14.11.1	for C and for the installed bore, unless OPTION FITTING_BASIS STANDARD
the same, above DN150	FM Global DS 3-0 Table 12(a)	as above
CPVC	the manufacturer's table	none

pipe type	table	correction
PE11, PE9	Le = K' x d (PPI Handbook of PE Pipe), d the actual bore	none
PVCM16	the steel tables as a controlled fallback	as for steel; the report states the assumption

Codes. Each table holds only some codes, and a code the table lacks for that pipe is an error. So is a size the table has no value for: a missing value is never taken as zero.

code	AS 2118.1 (DN150 and below)	FM DS 3-0 (above DN150)
E	90 deg standard elbow	standard screwed elbow, or run of tee reduced 1/2
EL	90 deg long radius elbow	90 deg long radius flanged elbow
E45	45 deg elbow	45 deg elbow
EF	-	90 deg flanged or medium sweep elbow, or tee reduced 1/4
T	tee or cross, flow turned 90 deg	standard tee or cross, flow turned 90 deg
GV	gate valve	gate valve, fully open
GV75 GV50 GV25	-	gate valve, 75 %, 50 %, 25 % open
CV	check or alarm valve (swing)	swing check, alarm check or dry pipe valve
CVM	check or alarm valve (mushroom)	-
CVB	check or alarm valve (butterfly)	-
BV	-	butterfly valve, fully open
AV	-	angle valve

code	CPVC
E	90 deg elbow
E45	45 deg elbow
T	tee, branch (flow turned 90 deg)
TR	tee, run: published, but not counted
CPL	coupling

code	PE (K')		code	PE (K')
E	90 deg moulded elbow (40)		E45M2	45 deg fabricated, 2+ mitres (15)
E45	45 deg moulded elbow (21)		E45M1	45 deg fabricated, 1 mitre (12)
E15	15 deg moulded elbow (6)		E30M2	30 deg fabricated, 2+ mitres (8)
EM3	90 deg fabricated, 3+ mitres (24)		E30M1	30 deg fabricated, 1 mitre (8)
EM2	90 deg fabricated, 2 mitres (30)		E15M1	15 deg fabricated, 1 mitre (6)
EM1	90 deg fabricated, 1 mitre (60)		T	tee, run to branch (60)
E60M2	60 deg fabricated, 2+ mitres (25)		TR	tee, run to run (20): not counted
E60M1	60 deg fabricated, 1 mitre (16)		GLV	globe valve (340)
AV	angle valve (145)		BV	butterfly valve, over 8 in (40)
CV	check valve, swing (135)			

The database is the authority. Its exact contents, and any changes made in the DATABASE tab, can be seen there. The editor's pick list offers the codes that exist for the pipe on the line.

11. Pipe types

The pipe types in the database as supplied. The DATABASE tab shows the current list, with every size's bore.

code	pipe	standard	sizes	C	roughness, mm
ASAM	AS 1074 medium steel tube	AS 1074	DN20-DN150	120	0.045
ASAH	AS 1074 heavy steel tube	AS 1074	DN20-DN150	120	0.045
S20	Australian Schedule 20 steel	ASME B36.10M	DN200-DN600	120	0.045
S40	Schedule 40 steel	ASME B36.10M	DN25-DN600	120	0.045
CPVC	BlazeMaster CPVC, SDR 13.5	ASTM F442	DN20-DN80	150	0.003
PE11	PE100, SDR 11, PN16	AS/NZS 4130	DN63-DN630	150	0.007
PE9	PE100, SDR 9, PN20	AS/NZS 4130	DN63-DN630	150	0.007

code	pipe	standard	sizes	C	roughness, mm
PVCM16	PVC-M, Series 2, PN16	AS/NZS 4765	DN100-DN450	150	0.003
BORE	not a type: the pipe is given by its bore		any	as USE gives, else 120	as USE gives, else 0.045 assumed

12. Names PHCP makes itself

A device in a pipe (BP, BFP, PRV, OPP, OFD, PRA, DPV) sits at the pipe's start. PHCP splits the pipe there:

- a new node `<pipe>~`, at the start node's elevation;
- the device, from the start node to `<pipe>~`, named `<pipe>.<command>` (for example `P1.BP`);
- the pipe itself, from `<pipe>~` to its end node.

These names appear in the report and the JSON. This is why `~` is reserved.

13. Messages

The reader stops at the first error and names its line. The editor shows every line's state as it is typed. The messages below are the commonest; the words in capitals are as PHCP prints them.

message	cause
UNKNOWN COMMAND X	The first word is not a command or a short form.
PIPE NEEDS SIZE	A field is missing. Each missing field is named.
'X': FIELDS GO BY POSITION - THERE ARE NO KEY=VALUE OPTIONS	A field contains `=`.
'!' CANNOT BE USED IN A .PHC LINE; COMMENTS START WITH //	A `!` or `#` outside a comment.
X IS ALREADY SET ON LINE N	A setting given twice.
NODE X IS ALREADY DEFINED ON LINE N	A node defined twice.
THE NAME X IS ALREADY USED ON LINE N	A pipe, pump or fixed flow name used twice.
NODE X IS NOT DEFINED	A node named but defined nowhere. Others are listed with it.
PIPE X ALREADY HAS A DEVICE, BP ON LINE N	A second device in one pipe.
UNKNOWN PIPE TYPE X (...)	Not in the database. The message lists the types.
DATA_REQUIRED: DN47 IS NOT IN THE DATABASE FOR PIPE TYPE ASAM	A size the type is not made in, or a fitting with no published value.
PIPE X: FITTINGS NEED A PIPE TYPE ...	Fittings on a USE BORE pipe. Use EQV.
UNKNOWN UNIT X FOR FLOW (...)	Not a unit of that quantity. The message lists them.
UNKNOWN OPTION X (...)	Not an option. The message lists them.
A PRESSURE RATIO VALVE REDUCES PRESSURE: ITS RATIO MUST BE MORE THAN 1	PRA ratio of 1 or less.
PUMPDUTY X: NO PUMP OF THAT NAME, AND NO PIPE WITH A BOOSTER (BP) IN IT	The duty names nothing it can check.
AREA A: X IS NOT AN OUTLET (SPRINKLER, HYDRANT OR HREEL)	An AREA names a plain node.
DESIGN_AREA X: NO AREA OF THAT NAME (...)	The option names an area the file does not have.
PIPE X: SIZE 0 ASKS FOR THE PIPE TO BE SIZED, WHICH NEEDS A PIPE TYPE ...	Auto sizing under USE BORE.
SIZING_MARGIN 'X' IS A PERCENTAGE OF THE SUPPLY (10%) OR A PRESSURE (50)	Not `10%` (under 100) or a pressure of 0 or more.
SIZING_TRIALS 'X' IS A WHOLE NUMBER OF TRIALS, 1 TO 50	Not a whole number from 1 to 50.
THE FILE HOLDS NO MODEL: NO NODES AND NO PIPES	Nothing but settings and comments.

Warnings do not stop the solve. They are listed in Section 12 of the report, in the editor's WARNINGS list and in the JSON. Among them: a pipe over VEL_LIMIT, an assumed roughness, a pump short of its duty, a ratio valve's inlet below its minimum, and a supply graph that could not be read.

14. A complete model

A single branch from a town main, through a backflow preventer, to four sprinklers and a hose reel. It is typed here in full, settings included. In the editor, the settings would be in the PROJECT tab and only the network lines typed.

```
// COMPLETE EXAMPLE FROM THE REFERENCE MANUAL
TITLE     REFERENCE MANUAL EXAMPLE
PROJECT   JOB HC-REF-001
UNITS     LENGTH M
UNITS     KFACTOR LPM/BAR
METHOD    HW
OPTION    HOSE_ALLOWANCE 250

// SUPPLY: TOWN MAIN, FLOW TEST 450 KPA STATIC
WATER     TOWN 0 450 1000 380 2000 250

// RISER AND MAIN
NODE      R0  0
NODE      R1  4
USE       ASAM
PIPE      M1  TOWN R0  30  100  GV
BFP       M1  0 10 1000 35 2000 60
PIPE      RS  R0  R1   4   100  E

// BRANCH LINE: FOUR K80 HEADS AT 100 KPA MINIMUM
SPRINKLER S1  4 -100 80
NC        S1  S2 S3 S4
PIPE      L1  R1  S1   3   65   T
PIPE      L2  S1  S2   3   50
PIPE      L3  S2  S3   3   40
PIPE      L4  S3  S4   3   32

// HOSE REEL OFF THE RISER, GIVEN BY BORE WITH AN ALLOWANCE
USE       BORE 120
HREEL     HR1 1  20  30
PIPE      HRP R0  HR1  5   27.3
EQV       HRP 2.5
```